The Seam Gate of the SHA-256 Message Schedule Universal Mixed-Core Geometry, Residue Quotients, Minimum Code Bounds, and Recoverability Across the Clean Echo Lattice

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Abstract

This paper develops the next structural layer above the clean echo intersection lattice of the SHA-256 message schedule. The clean echo program studies the negative space of the schedule recurrence: the expansion coordinates that a seed-word perturbation cannot contaminate before saturation locks. The prior lattice analysis showed that this negative space is highly structured. It separates into two parity-dual streams, an EVEN stream and an ODD stream, and the pairwise appearance frequency of every clean lattice node is controlled exactly by seed multiplicity $m(t)$ through the identity

$$f(t) = \frac{m(t)(m(t)-1)}{2}.$$

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The present paper proves that the MIXED part of that lattice contains a universal boundary core. Every MIXED pairwise intersection contains the same two-node seam:

$$K = \{16,17\}.$$

Node 16 is the first EVEN-stream coordinate. Node 17 is the first ODD-stream coordinate. Together they form the schedule's universal seam gate: a coupled boundary through which every two-parity clean crossing must pass. The seam is not empty. It is occupied by the crossing, but it is not privately owned by either parity stream.

Removing the universal seam from each MIXED intersection produces a finite seam-residue alphabet. The 36 MIXED seed-pair intersections collapse into exactly 11 residual classes. Therefore, any binary code that uniquely identifies the seam-residue class requires at least

$$\lceil \log_2 11 \rceil = 4$$

bits, while exact MIXED seed-pair identity requires at least

$$\lceil \log_2 36 \rceil = 6$$

bits. The paper therefore separates two different recoverability tasks: recovering crossing shape and recovering exact seed-pair identity. The seam stores class before it stores identity.

The result reframes the clean echo lattice as a storage geometry. Dirty coordinates are recurrence exhaust: they have already been reached by the contamination wave. Clean coordinates are stable support: they remain readable before saturation. The seam $K = \{16,17\}$ is the aperture of that aperture — the fixed boundary through which ODD and EVEN stream information becomes jointly readable.

Keywords: SHA-256, message schedule, clean echo set, seam gate, universal mixed core, residue quotient, parity stream, minimum code bound, recoverability, contamination recurrence, NEXUS framework, A-Mark9

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1. Introduction

The SHA-256 message schedule expands a 16-word input block into 64 schedule words. The recurrence is normally written as

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}}, \qquad 16 \le t \le 63.$$

The usual view is that this recurrence diffuses input information forward into later schedule words. That view is correct, but incomplete. A recurrence does not only define where influence goes. It also defines where influence does not go before the recurrence saturates. The clean echo program studies this negative space.

For each seed word $W[j]$, place a binary perturbation at j and ask which expansion coordinates $t \ge 16$ become contaminated by the recurrence. The complement of that contamination cone is the clean echo set C_j. These are the coordinates that remain unreachable by the perturbation before saturation locks the schedule into consecutive contamination.

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The prior clean echo lattice paper established that the family $\{C_j\}_{j=0}^{15}$ has a precise pairwise structure. It found:

1. A parity-dual stream structure:

$$S_E = \{16,18,20,22\},$$

$$S_O = \{17,19,21,23\}.$$

2. A complete pairwise census over the 120 unordered seed pairs:

24 ∅, 24 singleton, 21 ODD-only, 15 EVEN-only, 36 MIXED.

3. A node multiplicity law:

$$f(t) = \frac{m(t)(m(t)-1)}{2},$$

where $m(t)$ is the number of clean echo sets containing node t, and $f(t)$ is the number of seed-pair intersections in which t appears.

4. A lower ODD-stream seam $\{17,19\}$ that coincides with the ROTR constants of σ_1:

$$\sigma_1(x) = ROTR^{17}(x) \oplus ROTR^{19}(x) \oplus SHR^{10}(x).$$

The present paper focuses on the MIXED region. A MIXED intersection is one that contains at least one EVEN node and at least one ODD node. It is a pairwise crossing where both parity streams remain visible at once.

The main result is simple and exact:

Every MIXED intersection contains $\{16,17\}$.

This pair is the universal seam gate.

That result changes the interpretation of the clean echo lattice. The ODD and EVEN streams are not merely two lists of nodes. They are two reading channels. The MIXED class is not a vague overlap. It is a controlled boundary-crossing class. All such crossings pass through the same aperture:

$$S_E \mid K \mid S_O,$$

where

$$K = \{16,17\}.$$

The paper then asks the natural storage question:

What is the minimum information needed to specify a crossing through K?

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This is not a claim that SHA-256 is invertible. It is not a claim about breaking the compression function. It is a recurrence-level structural question about what clean information survives before saturation. The answer begins with the seam-residue alphabet.

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2. The Message Schedule as a Contamination System

2.1 The recurrence

The SHA-256 message schedule uses two small-sigma functions:

$$\sigma_0(x) = ROTR^7(x) \oplus ROTR^{18}(x) \oplus SHR^3(x),$$

$$\sigma_1(x) = ROTR^{17}(x) \oplus ROTR^{19}(x) \oplus SHR^{10}(x).$$

The schedule recurrence is

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}}.$$

For contamination geometry, the exact word values are not needed. The recurrence arms are what matter:

$$L = \{2,7,15,16\}.$$

A perturbation at one index can enter a later coordinate through any one of these lag arms.

2.2 Binary contamination trace

For a seed word $W[j]$, define a binary trace $B_t^{(j)}$. The value $B_t^{(j)} = 1$ means that coordinate t is contaminated by the perturbation seeded at j, while $B_t^{(j)} = 0$ means it remains clean.

Initialize

$$B_j^{(j)} = 1.$$

Then propagate by the recurrence arms:

$$B_t^{(j)} = B_{t-2}^{(j)} \vee B_{t-7}^{(j)} \vee B_{t-15}^{(j)} \vee B_{t-16}^{(j)}, \qquad t \geq 16.$$

The operator $\vee$ is Boolean OR. A coordinate becomes dirty if any prior arm feeding it is dirty.

2.3 Clean echo set

The clean echo set is the expansion-zone complement of the contamination trace:

$$C_j = \{t \in \{16, \ldots, 63\} : B_t^{(j)} = 0\}.$$

Thus C_j is not the path of contamination. It is the negative space left open by the contamination path.

This is the key inversion:

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dirty coordinates are exhaust; clean coordinates are stable readable residue.

Dirty coordinates have already been reached. Clean coordinates are still structurally distinguishable.

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3. Clean Echo Support

The clean echo sets used in this paper are the finite family produced by the SHA-256 lag set $L = \{2,7,15,16\}$ over the seed range $j = 0, \ldots, 15$.

Seed j	Direct-entry set $D(j)$	D(j)		Clean echo set C_j
0	{16}	1	{17, 19, 21}	22	PURE ODD
1	{16, 17}	2	Ø	16	ROOT-COVER / none
2	{17, 18}	2	{16}	17	PURE EVEN
3	{18, 19}	2	{16, 17}	18	MIXED
4	{19, 20}	2	{16, 17, 18}	19	MIXED
5	{20, 21}	2	{16, 17, 18, 19}	20	MIXED
6	{21, 22}	2	{16, 17, 18, 19, 20}	21	MIXED
7	{22, 23}	2	{16, 17, 18, 19, 20, 21}	22	MIXED
8	{23, 24}	2	{16, 17, 18, 19, 20, 21, 22}	23	MIXED
9	{16, 24, 25}	3	{17, 19, 21}	22	PURE ODD
10	{17, 25, 26}	3	{16, 18, 20, 22}	23	PURE EVEN
11	{18, 26, 27}	3	{16, 17, 19, 21, 23}	24	MIXED
12	{19, 27, 28}	3	{16, 17, 18, 20, 22, 24}	25	MIXED
13	{20, 28, 29}	3	{16, 17, 18, 19, 21, 23, 25}	26	MIXED
14	{16, 21, 29, 30}	4	{17, 19}	20	PURE ODD
15	{17, 22, 30, 31}	4	{16, 18, 20}	21	PURE EVEN

Two details matter for the present paper.

First, the stream nodes are the clean-support nodes that appear in at least two clean echo sets. These are the nodes that participate in pairwise intersections:

$$S_E = \{16,18,20,22\},$$

$$S_O = \{17,19,21,23\}.$$

Second, nodes 24 and 25 are clean-support nodes but not pairwise lattice nodes:

$$m(24) = m(25) = 1.$$

They appear in only one clean echo set each, so they do not appear in any pairwise intersection. This distinction is important: the seam gate belongs to the pairwise lattice, not merely to singleton clean support.

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4. Pairwise Intersections and Stream Classes

For each unordered seed pair (i, j) with $0 \le i < j \le 15$, define

$$I_{ij} = C_i \cap C_j.$$

Each I_{ij} falls into one of five operational classes:

1. **EMPTY:** $I_{ij} = \varnothing$.
2. **SINGLETON:** $|I_{ij}| = 1$.
3. **ODD-only:** all nodes in I_{ij} are odd.
4. **EVEN-only:** all nodes in I_{ij} are even.
5. **MIXED:** I_{ij} contains both parities.

The total number of unordered seed pairs is

$$\binom{16}{2} = 120.$$

The complete census is:

Class	Count	Meaning
EMPTY	24	No shared clean echo coordinate
SINGLETON	24	Exactly one shared clean coordinate
ODD-only	21	Shared clean residue only on ODD stream
EVEN-only	15	Shared clean residue only on EVEN stream
MIXED	36	Shared clean residue crosses both parity streams
Total	120	All unordered seed pairs

The seam gate lives inside the MIXED class.

5. Node Multiplicity and Pair Frequency

Before isolating the seam, it is useful to restate the pair-frequency law from the clean echo lattice.

For a node t, define seed multiplicity:

$$m(t) = |\{j : t \in C_j\}|.$$

If t appears in $m(t)$ clean echo sets, then the number of unordered seed pairs sharing t is

$$f(t) = \binom{m(t)}{2} = \frac{m(t)(m(t)-1)}{2}.$$

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This is not a probabilistic rule. It is a counting identity. Every pair of seed sets that contains t contributes one pairwise intersection containing t.

Node t	Seed multiplicity $m(t)$	Pair frequency $f(t)$	Parity stream	Pairwise status
16	12	66	EVEN	pairwise node
17	12	66	ODD	pairwise node
18	9	36	EVEN	pairwise node
19	9	36	ODD	pairwise node
20	6	15	EVEN	pairwise node
21	6	15	ODD	pairwise node
22	3	3	EVEN	pairwise node
23	2	1	ODD	pairwise node
24	1	0	EVEN	singleton support
25	1	0	ODD	singleton support

Nodes 16 and 17 are co-maximum:

$$m(16) = m(17) = 12,$$

so

$$f(16) = f(17) = \binom{12}{2} = 66.$$

This is the boundary-frequency seam. The universal MIXED seam will be the same pair:

$$K = \{16,17\}.$$

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6. The Seam Gate

6.1 Definition

The seam gate is

$$K = \{16,17\}.$$

Node 16 is the first EVEN-stream coordinate:

$$16 \in S_E.$$

Node 17 is the first ODD-stream coordinate:

$$17 \in S_O.$$

Together they form the first adjacent parity boundary in the expansion zone.

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This is the exact form of the separator:

$$S_E \mid K \mid S_O.$$

A more explicit display is

$$\{18,20,22\} \mid \{16,17\} \mid \{19,21,23\}.$$

The full streams include the seam endpoints:

$$S_E = \{16,18,20,22\},$$

$$S_O = \{17,19,21,23\}.$$

The seam is therefore not outside the streams. It is the shared boundary of the two stream systems.

6.2 Why this is a seam, not just a pair

A random shared pair would appear in some intersections and not others. The seam gate appears in every MIXED intersection. This gives it a stronger role.

It is not merely one signature among many. It is the invariant core of all two-parity crossings.

In ordinary set language:

$$K \subseteq I_{ij}$$

for every MIXED I_{ij}.

In operational language:

Any crossing that reads both parity channels must pass through $\{16,17\}$.

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7. Universal Mixed-Core Theorem

Theorem 7.1 — Universal Mixed-Core Theorem

Let $I_{ij} = C_i \cap C_j$ be a MIXED pairwise intersection. Then

$$\{16,17\} \subseteq I_{ij}.$$

Equivalently, for all MIXED pairs (i, j),

$$K \subseteq C_i \cap C_j.$$

Proof

The MIXED signatures in the complete pairwise lattice are exactly the following:

MIXED signature	Pair count

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MIXED signature	Pair count
{16, 17}	10
{16, 17, 18}	8
{16, 17, 19}	2
{16, 17, 18, 19}	5
{16, 17, 18, 20}	2
{16, 17, 19, 21}	2
{16, 17, 18, 19, 20}	2
{16, 17, 18, 19, 21}	2
{16, 17, 18, 20, 22}	1
{16, 17, 19, 21, 23}	1
{16, 17, 18, 19, 20, 21}	1

Each listed MIXED signature contains both 16 and 17. The list accounts for all 36 MIXED intersections. Therefore every MIXED intersection contains

$$K = \{16,17\}.$$

This proves the theorem. ▫

7.2 Interpretation of the proof

The proof is finite and exact because the SHA-256 seed range is finite: $0 \leq j \leq 15$. Once the recurrence is fixed, every C_j is determined. Once the C_j are determined, every I_{ij} is determined.

This is not a heuristic sample. It is an exhaustive enumeration over all 120 unordered seed pairs.

The theorem identifies the invariant part of the MIXED class. The variable part begins only after K is removed.

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8. Seam Residue Quotient

8.1 Definition

For each MIXED pair (i, j), define the seam residue

$$R_{ij} = I_{ij} \backslash K.$$

Since

$$K = \{16,17\},$$

this means

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$$R_{ij} = \big(C_i \cap C_j\big)\backslash\{16,17\}.$$

The map

$$I_{ij} \mapsto R_{ij}$$

factors out the universal mixed core and keeps only the extra clean residue attached to that crossing.

This is not a group quotient in the algebraic sense. It is a structural quotient: remove the invariant seam and classify what remains.

8.2 The residual alphabet

The 36 MIXED intersections produce exactly 11 seam-residue classes:

$$|\mathcal{R}_K| = 11.$$

The full alphabet is:

Seam residue R_{ij}	Count	Seed pairs
∅	10	(3,4), (3,5), (3,6), (3,7), (3,8), (3,11), (3,12), (3,13), (4,11), (11,12)
{18}	8	(4,5), (4,6), (4,7), (4,8), (4,12), (4,13), (5,12), (12,13)
{18, 19}	5	(5,6), (5,7), (5,8), (5,13), (6,13)
{18, 19, 20}	2	(6,7), (6,8)
{18, 19, 21}	2	(7,13), (8,13)
{18, 20}	2	(6,12), (7,12)
{19}	2	(5,11), (6,11)
{19, 21}	2	(7,11), (8,11)
{18, 19, 20, 21}	1	(7,8)
{18, 20, 22}	1	(8,12)
{19, 21, 23}	1	(11,13)

Theorem 8.1 — Seam Residue Alphabet Theorem

The seam-residue alphabet has exactly 11 classes:

$$|\mathcal{R}_K| = 11.$$

Proof

By Theorem 7.1, each MIXED signature contains $K = \{16,17\}$. Subtracting K from each of the 11 MIXED signatures gives the 11 residue classes listed above. The counts sum to

$$10 + 8 + 5 + 2 + 2 + 2 + 2 + 2 + 1 + 1 + 1 = 36,$$

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which equals the total number of MIXED pairs. Therefore the list is complete, and the residual alphabet has 11 classes. ▫

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9. Minimum Seam-Code Bounds

The seam-residue alphabet creates a storage bound.

A finite alphabet with N distinguishable symbols requires at least

$$\lceil \log_2 N \rceil$$

bits to encode uniquely.

Theorem 9.1 — Minimum Residue-Class Code Bound

Any binary code that uniquely identifies the seam-residue class of a MIXED intersection requires at least

$$\lceil \log_2 11 \rceil = 4$$

bits.

Proof

There are 11 distinct seam-residue classes. A b-bit binary code can represent at most 2^b classes. Therefore

$$2^b \geq 11.$$

Taking logarithms gives

$$b \geq \log_2 11.$$

Since b must be an integer,

$$b \geq \lceil \log_2 11 \rceil = 4.$$

▫

Theorem 9.2 — Minimum Exact-Pair Code Bound

Any binary code that uniquely identifies the exact MIXED seed pair requires at least

$$\lceil \log_2 36 \rceil = 6$$

bits.

Proof

There are 36 MIXED seed pairs. To identify the exact pair, a code must distinguish all 36 cases. A b-bit code can distinguish at most 2^b cases. Therefore

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$$2^b \geq 36,$$

so

$$b \geq \lceil \log_2 36 \rceil = 6.$$

▪

Corollary 9.1 — Shape Recovery and Identity Recovery Are Distinct

Seam-residue class recovery requires at least 4 bits. Exact seed-pair recovery requires at least 6 bits. Therefore

recovering the crossing shape is not the same as recovering the seed pair.

The seam stores crossing class before it stores identity.

This is the first-principles storage distinction opened by the clean echo lattice.

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10. Boundary Non-Ownership

The seam nodes are high-multiplicity addresses.

The seed-support set of node 16 is

$$\mathcal{S}(16) = \{j : 16 \in C_j\}$$

so

$$\mathcal{S}(16) = \{2,3,4,5,6,7,8,10,11,12,13,15\}.$$

The seed-support set of node 17 is

$$\mathcal{S}(17) = \{j : 17 \in C_j\}$$

so

$$\mathcal{S}(17) = \{0,3,4,5,6,7,8,9,11,12,13,14\}.$$

Both have multiplicity 12:

$$m(16) = 12,$$

$$m(17) = 12.$$

Their shared seed support is

$$\mathcal{S}(16) \cap \mathcal{S}(17) = \{3,4,5,6,7,8,11,12,13\}.$$

This means that 16 and 17 are not private nodes. Node 16 is EVEN, but its mixed behavior is coupled to node 17. Node 17 is ODD, but its mixed behavior is coupled to node 16.

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The seam is therefore best understood as non-owned boundary structure.

In Nexus language:

> unoccupiable does not mean empty; it means not privately ownable by either side.

The seam is occupied by the crossing. It is not owned by either stream.

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11. The Seam Decoder

The seam geometry gives a natural decoder.

For every seed pair (i, j):

1. Compute $I_{ij} = C_i \cap C_j$.
2. If $I_{ij} = \varnothing$, output **EMPTY**.
3. If $|I_{ij}| = 1$, output **SINGLETON**.
4. If every node in I_{ij} is odd, output **ODD-only**.
5. If every node in I_{ij} is even, output **EVEN-only**.
6. Otherwise output **MIXED**, and compute $R_{ij} = I_{ij} \backslash \{16,17\}$.

The decoder census is:

Decoder output	Count
EMPTY	24
SINGLETON	24
ODD-only	21
EVEN-only	15
MIXED with $R = \varnothing$	10
MIXED with $R \neq \varnothing$	26
Total	120

This decoder separates two kinds of mixed crossings.

A pure seam crossing has

$$R_{ij} = \varnothing,$$

so

$$I_{ij} = K = \{16,17\}.$$

A residue-bearing seam crossing has

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$$R_{ij} \neq \varnothing.$$

In that case the crossing carries additional channel information beyond the universal seam.

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12. Recoverability Across the Seam

12.1 The recoverability question

The seam-residue alphabet answers one question and opens another.

It answers:

> How many distinct mixed crossing shapes exist after the universal seam is removed?

Answer:

$$11.$$

It opens:

> What extra information is required to recover the exact seed pair?

Residue alone is not enough in general. Several residues occur for multiple seed pairs.

For example, $R = \varnothing$ occurs for 10 mixed pairs, while $R = \{18\}$ occurs for 8 mixed pairs. Thus residue identifies shape class, not identity.

12.2 Direct-entry count

The direct-entry count $|D(j)|$ is determined by when the seed j becomes eligible for each lag arm.

The recurrence arms are

$$L = \{2,7,15,16\}.$$

Seed j can enter the expansion zone through an arm once the corresponding target index is at least 16. This gives the threshold formula

$$|D(j)| = 1 + [j \geq 1] + [j \geq 9] + [j \geq 14],$$

where $[P]$ is 1 if proposition P is true and 0 otherwise.

Thus:

- $|D(0)| = 1$,
- $|D(1)| = \cdots = |D(8)| = 2$,
- $|D(9)| = \cdots = |D(13)| = 3$,
- $|D(14)| = |D(15)| = 4$.

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For a seed pair (i, j) define

$$d_{ij} = |D(i)| + |D(j)|.$$

The seam residues correlate with this value as follows:

Seam residue R_{ij}	Count	Range of $\|D(i)\| + \|D(j)\|$	Unique exact pair?	Example / first pairs
∅	10	4..6	no	(3,4), (3,5), (3,6), (3,7), (3,8) …
{18}	8	4..6	no	(4,5), (4,6), (4,7), (4,8), (4,12) …
{18, 19}	5	4..5	no	(5,6), (5,7), (5,8), (5,13), (6,13)
{18, 19, 20}	2	4	no	(6,7), (6,8)
{18, 19, 21}	2	5	no	(7,13), (8,13)
{18, 20}	2	5	no	(6,12), (7,12)
{19}	2	5	no	(5,11), (6,11)
{19, 21}	2	5	no	(7,11), (8,11)
{18, 19, 20, 21}	1	4	yes	(7,8)
{18, 20, 22}	1	5	yes	(8,12)
{19, 21, 23}	1	6	yes	(11,13)

Two singleton residue classes are especially informative:

$$R = \{18,19,20,21\}$$

occurs only for pair $(7,8)$, while

$$R = \{19,21,23\}$$

occurs only for pair $(11,13)$.

These residues already recover exact pair identity without additional side information.

Most residues do not. For them, side data is required. Candidate side data includes saturation thresholds S_i, S_j, direct-entry counts $|D(i)|, |D(j)|$, aperture class, seed neighborhood, and higher-order intersections involving a third clean echo set.

The recoverability problem is therefore not binary. It has levels:

residue class → reduced candidate pair set → exact pair.

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13. Storage Interpretation

The clean echo lattice is a storage map because clean coordinates are the coordinates that remain stable before saturation completes.

Dirty coordinates are not useless, but they are no longer clean carriers. They have already been touched by the contamination wave. In this analysis they are exhaust.

Clean coordinates are negative-space carriers. They tell us where the recurrence has not yet erased distinguishability.

The MIXED seam is the minimal two-channel storage boundary:

$$S_E \mid K \mid S_O.$$

The seam K is the boundary condition. The residue R_{ij} is the extra information riding on that boundary.

Thus the storage structure has three layers:

1. **Universal seam:** $K = \{16,17\}$.
2. **Residual shape:** $R_{ij} = I_{ij} \backslash K$.
3. **Exact pair identity:** (i, j).

These layers have different information costs:

$$K \quad \text{is invariant,}$$

$$R_{ij} \quad \text{requires 4 bits to classify,}$$

$$(i, j) \quad \text{requires 6 bits among MIXED pairs.}$$

This is the main structural conclusion of the paper.

---

14. Relation to the σ_1 Fingerprint

The previous clean echo lattice paper identified the lower ODD-stream seam $\{17,19\}$ as the integer-label match to the ROTR constants of

$$\sigma_1(x) = ROTR^{17}(x) \oplus ROTR^{19}(x) \oplus SHR^{10}(x).$$

The present paper adds another role for node 17.

Node 17 is:

1. the first ODD-stream node after the expansion boundary,
2. the lower ROTR constant of σ_1,
3. the ODD half of the universal MIXED seam $K = \{16,17\}$.

So 17 sits at the intersection of two structures:

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$$\text{boundary-frequency seam} = \{16,17\},$$

$$\sigma_1 \text{ fingerprint seam} = \{17,19\}.$$

Their overlap is

$$\{16,17\} \cap \{17,19\} = \{17\}.$$

This does not prove causal selection of the σ_1 constants. The paper retains the fingerprint-level claim only. A stronger selection claim requires a formal coupling map from schedule-time coordinates to bit-rotation transport geometry.

But the structural role of 17 is now sharper:

$$\boxed{17 = \text{boundary hinge between seam storage and ODD-stream fingerprinting.}}$$

---

15. Broader Two-Channel Mold

The seam-gate result has the form:

$$\boxed{\text{two channels} + \text{fixed boundary} + \text{residual alphabet.}}$$

This form appears in many systems, but the present paper proves it only for the SHA-256 clean echo lattice. The following comparisons are structural correspondences, not proofs of identity.

15.1 Complementary rails

A two-channel separator resembles any system where two rails carry complementary states while a boundary rule determines admissible crossings. DNA strand pairing is one familiar example: two complementary rails, constrained pairing, and a readable seam between them.

The SHA result is not biochemical. The point is the mold:

$$\text{rail}_0 \mid \text{seam} \mid \text{rail}_1.$$

15.2 Coprime / non-coprime wheel split

In the clean echo lattice, the ODD-stream nodes $17,19,23$ are coprime to $210 = 2 \cdot 3 \cdot 5 \cdot 7$, while $21 = 3 \cdot 7$ is not. The EVEN stream nodes $16,18,20,22$ are all non-coprime to 210 because each is even.

This creates a wheel-like split:

$$\text{coprime-facing ODD residues} \quad \text{vs.} \quad \text{non-coprime EVEN residues.}$$

Again, the present paper does not claim the wheel causes the schedule. It records the matching mold.

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15.3 Sziklai-style conjugate faces

The Sziklai identity in SHA-256 round analysis exposes a conserved relation between two faces of the compression update. The seam gate has a similar abstract shape: two faces, one crossing, and residue that survives the update boundary.

The connection remains a formal analogy until a direct operator map is written.

---

16. Complete Mixed-Pair Appendix

The following table lists every MIXED pair, its full intersection, and its seam residue after removing $K = \{16,17\}$.

Pair (i,j)	Full MIXED intersection I_{ij}	Seam residue R_{ij}	Residue size
(3,4)	{16, 17}	∅	0
(3,5)	{16, 17}	∅	0
(3,6)	{16, 17}	∅	0
(3,7)	{16, 17}	∅	0
(3,8)	{16, 17}	∅	0
(3,11)	{16, 17}	∅	0
(3,12)	{16, 17}	∅	0
(3,13)	{16, 17}	∅	0
(4,5)	{16, 17, 18}	{18}	1
(4,6)	{16, 17, 18}	{18}	1
(4,7)	{16, 17, 18}	{18}	1
(4,8)	{16, 17, 18}	{18}	1
(4,11)	{16, 17}	∅	0
(4,12)	{16, 17, 18}	{18}	1
(4,13)	{16, 17, 18}	{18}	1
(5,6)	{16, 17, 18, 19}	{18, 19}	2
(5,7)	{16, 17, 18, 19}	{18, 19}	2
(5,8)	{16, 17, 18, 19}	{18, 19}	2
(5,11)	{16, 17, 19}	{19}	1
(5,12)	{16, 17, 18}	{18}	1
(5,13)	{16, 17, 18, 19}	{18, 19}	2
(6,7)	{16, 17, 18, 19, 20}	{18, 19, 20}	3

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Pair (i, j)	Full MIXED intersection I_{ij}	Seam residue R_{ij}	Residue size
(6,8)	{16, 17, 18, 19, 20}	{18, 19, 20}	3
(6,11)	{16, 17, 19}	{19}	1
(6,12)	{16, 17, 18, 20}	{18, 20}	2
(6,13)	{16, 17, 18, 19}	{18, 19}	2
(7,8)	{16, 17, 18, 19, 20, 21}	{18, 19, 20, 21}	4
(7,11)	{16, 17, 19, 21}	{19, 21}	2
(7,12)	{16, 17, 18, 20}	{18, 20}	2
(7,13)	{16, 17, 18, 19, 21}	{18, 19, 21}	3
(8,11)	{16, 17, 19, 21}	{19, 21}	2
(8,12)	{16, 17, 18, 20, 22}	{18, 20, 22}	3
(8,13)	{16, 17, 18, 19, 21}	{18, 19, 21}	3
(11,12)	{16, 17}	∅	0
(11,13)	{16, 17, 19, 21, 23}	{19, 21, 23}	3
(12,13)	{16, 17, 18}	{18}	1

17. Reproducibility

The following Python code reproduces the clean echo sets, pairwise classifications, and seam residue alphabet.

```
from collections import Counter, defaultdict
from math import ceil, log2

N = 64
K = frozenset({16, 17})

def clean_echo(j, n=N):
  B = [0] * n
  B[j] = 1
  for t in range(16, n):
    B[t] = int(B[t-2] or B[t-7] or B[t-15] or B[t-16])
  return frozenset(t for t in range(16, n) if B[t] == 0)

C = {j: clean_echo(j) for j in range(16)}

classes = Counter()
residues = Counter()
```

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```
residue_pairs = defaultdict(list)

for i in range(16):
  for j in range(i + 1, 16):
    I = C[i] & C[j]
    if len(I) == 0:
      classes['EMPTY'] += 1
    elif len(I) == 1:
      classes['SINGLETON'] += 1
    elif all(t % 2 == 1 for t in I):
      classes['ODD-only'] += 1
    elif all(t % 2 == 0 for t in I):
      classes['EVEN-only'] += 1
    else:
      classes['MIXED'] += 1
      assert K.issubset(I)
      R = frozenset(I - K)
      residues[R] += 1
      residue_pairs[R].append((i, j))

print(classes)
print(residues)
print('residue bits:', ceil(log2(len(residues))))
print('exact mixed-pair bits:', ceil(log2(classes['MIXED'])))
```

Expected pairwise census:

$$24 \text{ EMPTY}, \quad 24 \text{ SINGLETON}, \quad 21 \text{ ODD-only}, \quad 15 \text{ EVEN-only}, \quad 36 \text{ MIXED}.$$

Expected seam result:

$$|\mathcal{R}_K| = 11.$$

Expected code bounds:

$$\lceil \log_2 11 \rceil = 4,$$

$$\lceil \log_2 36 \rceil = 6.$$

---

18. Open Problems

18.1 Formal coupling map

The σ_1 fingerprint remains an integer-label coincidence until a formal coupling map is supplied between schedule-time coordinates and bit-rotation depths.

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The target is to explain why $\{17,19\} \subset S_O$ also appears as $\{ROTR17, ROTR19\}$ inside σ_1.

18.2 Exact pair recovery from residue and side data

The seam residue alphabet identifies 11 crossing classes, not all 36 exact MIXED pairs. What extra invariant is needed to lift

$$R_{ij} \mapsto (i, j)?$$

Candidate side data includes S_i, S_j, $|D(i)|, |D(j)|$, aperture class, seed neighborhood, and higher-order intersections.

18.3 Higher-order seam intersections

The present paper studies pairwise intersections. The next object is

$$\bigcap_{j \in J} C_j.$$

Questions:

1. What is the highest-order mixed seam?
2. Does every higher-order mixed seam still factor through $K = \{16,17\}$?
3. Is there a closed-form hierarchy for seam residues at order k?

18.4 General lag sets

For a general recurrence lag set $L = \{\ell_1, \dots, \ell_k\}$, when does a universal mixed seam exist?

A candidate condition is the presence of one parity-preserving arm, one parity-bridging arm, and adjacent boundary activation at the expansion threshold. SHA-256 has exactly this shape through the $t - 2$ arm and the $t - 7$ bridge.

18.5 Cryptanalytic relevance

The seam gate is a schedule-level structure. It does not automatically survive the nonlinear compression function.

A practical experiment would perturb seed pairs sharing the same seam residue and measure whether downstream compression traces cluster by residual class. If clustering exists, seam residue may be visible beyond the schedule. If not, the compression function reseals the schedule-level seam.

18.6 Transformer attention transport hypothesis

The clean echo papers have suggested that schedule null zones may have analogues in attention systems as stable perturbation-null positions. A seam test would ask whether two-channel attention flows have universal boundary tokens analogous to $K = \{16,17\}$.

This remains speculative until experimentally tested.

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19. Conclusion

The clean echo lattice reveals where information survives in the SHA-256 message schedule before saturation completes. The present paper identifies the universal boundary of that survival geometry:

$$K = \{16,17\}.$$

Every MIXED pairwise intersection contains this seam. Therefore every two-parity crossing in the clean echo lattice passes through the same fixed boundary.

Removing the seam produces the seam-residue alphabet:

$$R_{ij} = I_{ij} \backslash K.$$

The 36 MIXED crossings collapse into 11 residual classes:

$$36 \to 11.$$

This gives the first minimum code bounds:

$$\text{residue-class recovery} = 4 \text{ bits minimum},$$

$$\text{exact mixed-pair recovery} = 6 \text{ bits minimum}.$$

The result separates crossing-shape recovery from exact seed-pair recovery. The seam stores class before identity.

The final structure is:

$$\boxed{S_E \mid K \mid S_O}$$

with

$$S_E = \{16,18,20,22\},$$

$$K = \{16,17\},$$

$$S_O = \{17,19,21,23\}.$$

This is the seam gate of the SHA-256 message schedule: a minimum-resistance two-channel separator with a fixed boundary and a finite residual alphabet.

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